Contact Modelling

Athul C D
Mechanical Engineering
IIT Bombay
Mumbai, India
athulcd@iitb.ac.in

***Abstract*—This report summarises the work done for the project conjunction with the course ME 472: Nonlinear Dynamics. This report explores various contact models present in the literature with emphasis on the compliant nonlinear contact models. Various widely used models for dynamic real-time simulations are compared analytically and physically. Various parameters of performance like coefficient of restitution, impact response etc. This report also discusses the current state-of-art in contact modelling research while also having a look into the possibilities for future release.**

***Index Terms*—Contact Modelling, Nonlinear Dynamics**

I. Introduction

Impact is a complex phenomenon that occurs when two or more bodies undergo a collision. Contact is a more ambiguous term although it is frequently used interchangeably with impact. It is used to describe situations where two or more bodies come in touch with each other at some locations. Inherently, contact implies a continuous process which takes place over a finite time. To model contact we study the coefficient of restitution, deformation/penetration of the surface which is being impacted upon, the forces and energy involved etc

Contact modelling is essential for robotic applications, Machine design and in general for elements which regularly makes contact with environment while performing routine tasks like finger in contact with an object while grasping. Modeling of contact is thus an important aspect of dynamic simulation of such systems. This report considers a computationally efficient compliant contact model with nonlinear damping. Computational simplicity is very important because it makes it possible for dynamic simulations which are performed real-time. Rigid body approximation provides a simplified model of real objects which are compliant. Modelling Compliance helps us to identify when the compliant approximation is a valid.

II. Earlier Work

Many models for modelling contact exist in the literature [5]. The model I decided to study was a Compliant Contact Model with Nonlinear Damping, which was introduced late in 90s and grew in popularity in 2000s. Hertz contact model was one of the earliest models for modelling contact. In his model elastic theory was used to calculate local indentation without the use of damping. It is given by the expression-

$$F_n = K\delta^n$$

here k and n are constants, depending on material and geometric properties. Although nonlinear, this model doesn't take damping into account. Experimentally it was was found that this model can be used with reasonable accuracy for low impact speeds and hard materials and fails to give accurate values in other cases.

Kelvin-Voigt Model is one of the first models to incorporate viscoelasticity. It can be represented by a purely viscous damper and purely elastic spring connected in parallel. It can

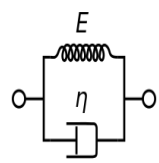

Fig. 1. Schematic of Kelvin-Voigt model

be represented by the expression

$$\sigma(t) = E\epsilon(t) + \eta\frac{d\epsilon(t)}{dt}$$

This model is inconsistent with some of the initial/boundary conditions. The contact force at the beginning of impact is discontinuous, because of the non-zero damping term, which is present even at the moment of impact. In a more realistic model, both elastic and damping forces should be initially zero and increase over time.As the objects are separating(indentation tends to zero), their relative velocity tends to be negative. As a result, a negative force holding the objects together is present. Again this is not physically consistent. The equivalent coefficient of restitution(e) defined for this model does not depend on impact velocity. But velocity dependence of e has been demonstrated experimentally and empirical relations relating v and e exist in the literature.

III. Nonlinear Contact Model

Since the linear Kelvin-Voigt model fails to model the physical intial/boundary conditions [1] tries to propose a compliant contact model with nonlinear damping. The model should be computationally simple enough to be run dynamically in a system with multiple contacts. It should be able to handle impact, sustained contact under load, and transition to and from contact conditions. It should be applicable to a variety of contact situations, rather than limiting its use to rebounding impacts.

Two major forces appear during contact: the normal force and the friction force. The work done in [1] focusses mainly on the normal force. Their model is given by

$$f = -(\lambda x^n)\dot{x} - kx^n$$

Penetration - x
penetration velocity - $\dot{x}$
damping constant - λ
spring constant - k
power - n (often close to 1 and depends on the surface geometry of the contact)
For hertzian contact of a sphere it is known that n= $\frac{3}{2}$

IV. Impact Response

A. Mathematical formulation

The empirical relation $e = 1 - \alpha v_i$ can be obtained by mathematically solving the system. This holds for $\lambda = \frac{3}{2}\alpha k$. The contact force has no discontinuities at initial contact and separation, but it begins and finishes with the correct value of zero.

$$f = m\ddot{x} = -\lambda x^n \dot{x} - kx^n$$

$$\ddot{x} = -\Lambda x^n \dot{x} - Kx^n$$

where $\Lambda = -\frac{\lambda}{m}$ and K = $-\frac{k}{m}$ It can also be written as

$$\dot{x} = y$$

$$\dot{y} = \Lambda y x^n + Kx^n$$

Fixed point (0,0)
x-nullcline y = 0
y-nullcline x = 0, y = -$\frac{2}{3\alpha}$

$$\frac{dv}{dx} = \frac{\dot{v}}{\dot{x}} = \frac{\Lambda x^n v + Kx^n}{v} = \frac{(\Lambda v + K)x^n}{v}$$

$$\int \frac{vdv}{\Lambda v + K} = \int x^n dx$$

V. Performance Characteristics

A. Phase Diagram

Each contour on the diagram shows how the position and velocity of the mass will vary for a given initial conditions on the contour and no external forces. For an impact, the contour is entered at x=0 and equal to a positive initial velocity, v_i , which causes a positive penetration, The mass follows the contour back around to x=0. The system forms periodic orbits centered at origin, but from a physical point-of-view only the right half plane makes sense. Left hand side corresponds to negative penetration, which is meaningless. The contours converge to a fixed separation velocity as the initial penetration velocity becomes large. This corresponds to the neighbourhood of th y-nullcline $\frac{2}{3\alpha}$ where $\dot{x}$ is nearly zero. A more intuitive/physical reason for this is that when the velocity reaches v= -$\frac{2}{3\alpha}$ the spring and damper forces exactly cancel. This causes the mass to return to the surface at that constant velocity since there are no more forces acting on it. If

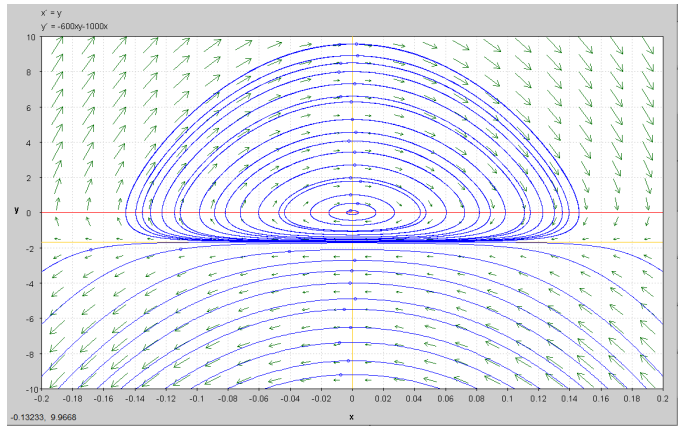

Fig. 2. Phase portrait for m = 50 kg, k = 50 kN/m, n = 1 $\alpha = 0.4$

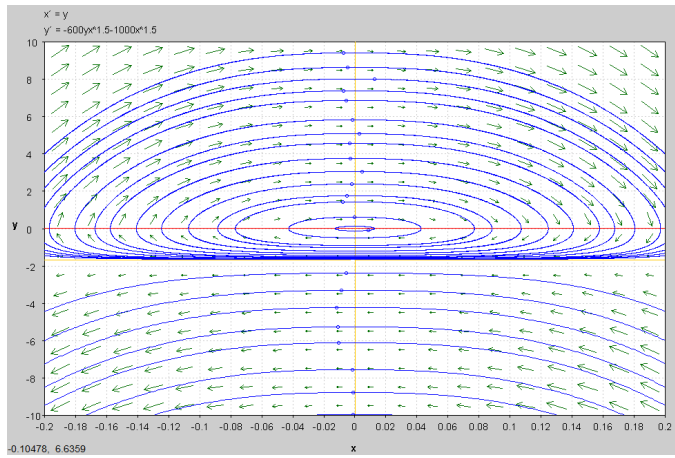

Fig. 3. Phase portrait for m = 50 kg, k = 50 kN/m, n = 1.5 $\alpha = 0.4$

particle impacts the system with a positive velocity it can never attain a velocity below -$\frac{2}{3\alpha}$ System can get in to this contours only when an external force is actively trying to separate the objects. The system is still stable and lossy if it gets on one of these contours because the trajectory still takes it back to x= 0 with a decreasing velocity.

B. Coefficient of Restitution

Empirically it is found that coefficient of restitution $e = 1 - \alpha v_i$, for low velocities. Value of e can be calculated from the exact solution of the second order differential equation or using the phase portrait. The model proposed follows this empirical relation for low values of v and for higher values of v it leads to a previously unknown result. The plot of coefficient of restitution v/s log of impact velocity gives a constant slope. It has been proved experimentally by [4] that it is indeed a correct observation.

VI. Current State of Art

Most of the models which commonly appear in literature is of the form f = $kz^a + \lambda z^b \dot{z}^c$. A lot of work has been done

to obtain combinations (a,b,c) such that model can predict the contact in a better way.

$$F = kz^{\frac{3}{2}} + \lambda z^{\frac{1}{2}}\dot{z}$$

A model proposed by [4] uses $a = \frac{3}{2}, b = \frac{1}{2}, c = 1$. Their experiments have found out that this gives a better result experimentally compared to the one proposed by [1] In the last few years reinforcement learning is also widely used to model and simulate contact in robotics specially in problems of grasping and lifting objects using finger like mechanisms. But it's notorious for the huge computational time involved and is difficult to implement it for real-time simulations. Also it involves tuning a lot of of parameters.

VII. Possibilities for research

The models covered in this paper and the ones in literature focus primarily on modelling normal force. Modelling of friction can be done and a more comprehensive model which deals with friction can be developed. From the perspective of energy we can modify our model to account for the energy lost. Energy lost via wave propagation, heat and sound, plasticity etc can be incorporated into the model. The non linear component of the damping and elastic term can be modelled in a different way. Currently all models are of the form f = $kz^a + \lambda z^b \dot{z}^c$
Contact between Polyhedral Objects can be studied similar to that of sphere and attempts can be made to get a relationship between the geometry of the body and exponent n. Body undergoing multiple contacts simultaneously can be studied further with suitable additions to existing models.

VIII. Conclusion

As per the compliant contact model with nonlinear damping proposed in [1] coefficient of restitution has now been characterized over an entire range of impact velocities and found to vary in a physically meaningful manner with damping and impact velocity. The model is lossy and thus stable under all initial conditions, and that sticky tensile forces do not occur in simple impacts. The model is simple enough to be implemented for dynamic simulations and is widely used in robotics, machine design, multi-body analysis. This has lead to widespread adoption of the models in problems involving real-time simulation.

IX. Acknowledgement

I would like to Prof Dyanesh Pawaskar for providing the opportunity to work on this project.

X. Supplementary Information

The phase portrait was generated using PPlane (https://math.rice.edu/ dfield/dfpp.html)

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